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Complete Research Methodology

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Deakin University

SIT746 Research Project (Advanced)

Complete Research Methodology

Detection and Simulation of GPS Spoofing in Autonomous Mobile Robots Using Sensor Cross-Validation

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Introduction

Autonomous mobile robots such as unmanned aerial vehicles (UAVs), ground robots, and other embedded robotic platforms depend heavily on GPS and broader GNSS services for localisation, path following, waypoint navigation, and autonomous mission execution. Civilian GPS is, however, structurally vulnerable because its signals are weak, publicly documented, and generally unauthenticated at the receiver level, making spoofing a realistic threat to robotic navigation systems [1–3].

GPS spoofing is particularly dangerous in autonomous robots because the navigation estimate is directly coupled to estimation, planning, and control. A spoofed position can therefore produce not only localisation error but also unsafe control behaviour, mission deviation, and loss of system trustworthiness in cyber-physical environments [3, 4]. This makes GPS spoofing in robotics not merely a radio-frequency anomaly problem, but a systems security problem spanning sensing, inference, control, and resilience.

This methodology document formalises the research approach initiated in SIT723 and continued in SIT724/SIT746. The project has already established a PX4–Gazebo simulation environment, a MAVLink-based telemetry capture pipeline, automated data processing, sliding-window dataset creation, and baseline machine-learning-based anomaly detection infrastructure, and this document now presents the formal research methodology that guides the remaining stages of the work [11–13].

The methodology is designed as a work-in-progress research framework rather than a claim of completed final validation. In particular, the methodology formalises how the project will move from baseline telemetry anomaly detection to full cross-sensor validation, continuous GPS trust assessment, and quantitative breaking-point analysis under controlled spoofing scenarios [11, 12].

Research Problem, Aim, and Questions

Research Problem

Existing literature shows that GPS spoofing detection is possible using signal-level monitoring, sensor fusion, machine learning, and simulation-based analysis. However, important gaps remain for autonomous mobile robots: platform-specific vulnerability analysis is limited; many real-time sensor fusion methods are not validated under practical robotic constraints; and most detection systems still output binary spoofed/authentic decisions instead of continuous trust estimates that can support graded navigation responses [1, 3, 5, 9].

In practical PX4-class and similar embedded robotic environments, these gaps are amplified by limited access to low-level receiver internals, finite compute resources, real-time latency

constraints, and the need to distinguish spoofing from normal dynamic flight disturbances. The research problem addressed by this project is therefore how to design a practical, reproducible, and real-time GPS spoofing detection methodology that is suitable for autonomous mobile robots and grounded in telemetry that is realistically available in deployed robotic systems [3, 6, 7].

Research Aim

The aim of this research is to develop and evaluate a real-time GPS spoofing detection methodology for autonomous mobile robots using sensor cross-validation, machine learning, and dynamic trust assessment within a reproducible PX4–Gazebo simulation environment.

Research Questions

The methodology is guided by the following research questions:

- RQ1:** What are the primary vulnerabilities of GPS in autonomous mobile robots to spoofing attacks?
- RQ2:** How can sensor fusion techniques effectively detect GPS spoofing in real time?
- RQ3:** How can a dynamic GPS Trust Index be designed to quantify the reliability of GPS data?

Research Objectives

To answer the research questions, the project has the following objectives:

1. To analyse platform-specific GPS spoofing vulnerabilities in autonomous mobile robots.
2. To collect and preprocess real-time telemetry from PX4-based simulation through MAVLink.
3. To construct labelled datasets representing normal and spoofed navigation behaviour.
4. To develop and compare suitable machine learning approaches for real-time anomaly detection.
5. To implement sensor cross-validation between GPS and non-GPS telemetry sources such as IMU- and estimator-related measurements.
6. To design a Dynamic GPS Trust Index that converts anomaly evidence into a continuous reliability estimate.
7. To quantify the operational limits or breaking point at which spoofing detection performance begins to degrade.

Research Design

Overall Research Approach

This research adopts an experimental, design-oriented, and data-driven methodology. It is experimental because spoofing scenarios are generated and evaluated under controlled conditions. It is design-oriented because the project develops a working detection framework rather than only analysing existing theory. It is data-driven because anomaly detection and trust estimation are based on telemetry collection, feature extraction, labelled datasets, and model evaluation [11, 13].

The methodology also includes iterative refinement. Earlier stages of the project established the simulation environment, telemetry pipeline, baseline dataset generation, and initial model training; the current methodology extends that foundation into full research design, formal evaluation, risk management, ethical governance, and dissemination planning [12, 13].

Research Setting

The primary experimental setting is a PX4 Software-in-the-Loop (SITL) environment integrated with Gazebo and external telemetry monitoring via MAVLink. This setup was selected because it supports safe, repeatable, and controllable spoofing experimentation while exposing the kinds of telemetry interfaces that are realistic for robotic autopilot systems [7, 8, 11, 12].

The current methodology prioritises simulation-driven evaluation because controlled GPS spoofing on live physical systems raises safety, legal, and technical constraints. Simulation therefore serves as the main research environment for data generation, attack injection, detection development, and breaking-point analysis, with any later hardware extension treated as future work rather than a current methodological requirement [7, 8, 12].

Conceptual Framework

The conceptual structure of the methodology aligns directly with the three research questions:

- **RQ1 — GPS Vulnerabilities:** identifies the threat model and platform-specific attack surface.
- **RQ2 — Sensor Fusion Detection:** develops real-time detection using GPS, IMU-related, estimator, and system-health telemetry.
- **RQ3 — Trust Index:** transforms anomaly evidence into a continuous GPS reliability score for navigation-aware decision support.

These three components are integrated into a single detection framework in which vulnerability analysis motivates the feature and attack design, sensor fusion provides the core detection

mechanism, and trust assessment provides the decision-level interpretation of the results.

Overview of Research Methods

Candidate Research Methods in This Topic

Several research methods are relevant to GPS spoofing detection in autonomous mobile robots:

1. **Signal-level detection:** power monitoring, signal quality monitoring, AGC analysis, and correlator-based analysis.
2. **Rule-based anomaly detection:** heuristic thresholds applied to telemetry or navigation consistency measures.
3. **Sensor fusion methods:** cross-validation using GPS, IMU, barometer, optical flow, or estimator outputs.
4. **Machine learning methods:** supervised anomaly detection using feature vectors or time-series windows.
5. **Simulation-based experimental evaluation:** controlled scenario generation using robotic simulation frameworks.

Strengths, Weaknesses, and Suitability

Table 1: Comparison of Research Methods Relevant to This Project

Method	Strengths	Weaknesses	Suitability
Signal-level detection	Directly targets spoofing indicators at source level; can detect simple attacks early	Often requires receiver internals unavailable in COTS robotic systems	Moderate
Rule-based telemetry detection	Interpretable, fast, easy to prototype, useful for weak supervision	Sensitive to threshold design; can overfit to known cases	High as baseline
Sensor fusion	Uses physical consistency across sensors; well aligned with robotic state estimation	Depends on sensor availability and synchronisation; can degrade under noisy motion	Very High
Machine learning	Learns nonlinear spoofing patterns and can improve robustness over simple rules	Needs labelled data and careful validation; risk of overfitting	Very High
Simulation-based evaluation	Safe, reproducible, controlled, scalable for scenario generation	Sim-to-real gap; may not capture all hardware noise effects	Very High

Chosen Methodology

The selected methodology combines four elements:

1. controlled spoofing scenario generation in PX4–Gazebo simulation,
2. MAVLink telemetry collection and structured preprocessing,
3. rule-based weak supervision and feature engineering for labelled datasets,
4. machine learning and sensor cross-validation for real-time anomaly detection and trust assessment.

This combined method is selected because it aligns well with all three research questions. It supports vulnerability analysis through controlled attack design, supports real-time sensor fusion detection through telemetry-level modelling, and supports trust-index design through probability-driven reliability scoring [11–13].

Justification for Method Choice

This methodology is likely to produce valid results because it evaluates the detector under controlled and repeatable spoofing conditions, uses explicit preprocessing and feature engineering rules, and measures performance through standard classification and latency metrics. It is also reliable because the same pipeline can be rerun across multiple logs and scenarios using scripted, configuration-driven processing [11].

The chosen approach is especially appropriate for this project because it matches the actual research infrastructure already built: the project currently includes a working PX4–Gazebo setup, MAVLink telemetry logging, automated cleaning and labelling, window-based model training, live inference, and a monitoring dashboard [11, 12]. This makes the methodology both academically defensible and practically executable within the scope of the unit.

System Architecture and Implementation Context

Architecture Overview

The current system architecture consists of the following components:

1. **PX4 SITL / Gazebo Simulation Layer** for mission execution and attack scenario generation.
2. **MAVLink Telemetry Layer** for streaming GPS and related telemetry data.
3. **Data Processing Pipeline** for cleaning, auto-labelling, window creation, and feature engineering.
4. **Machine Learning Layer** for anomaly detection using trained models.

5. **Live Inference Layer** for real-time detection from streaming telemetry.

6. **Visualisation Layer** through a Streamlit dashboard for monitoring and replay analysis.

This architecture emerged partly from a technical change in the PX4 toolchain that reduced direct container access and forced a move toward a MAVLink UDP bridge and external Python-based processing. That architectural change is now embedded into the methodology because it more closely resembles a realistic companion-computer deployment model for robotic systems [12].

Current Pipeline Status

The project documentation shows that the telemetry-driven pipeline already supports raw log collection, automated data cleaning, auto-labelling, window generation, feature engineering, model training, and live inference operation. The methodology therefore builds on a working implementation rather than a purely conceptual design [11, 12].

At the same time, the methodology remains honest about staged completion. Checkpoint documentation identified several components as still requiring formal completion or validation, including full cross-sensor algorithm implementation, anomaly threshold tuning, real-time lag quantification, and breaking-point measurement as a core research result [12]. These are therefore treated as formal methodological tasks within the research plan rather than as completed final outcomes.

Data Collection Methodology

Telemetry Source

Data collection is performed through MAVLink over UDP, using the PX4 SITL environment as the telemetry source. This method was selected because it captures realistic autopilot-level navigation and system information without requiring low-level GNSS receiver internals that are often inaccessible in commercial robotic systems [7, 11, 12].

Captured Message Types

The telemetry pipeline captures multiple MAVLink message types, including:

- GLOBAL_POSITION_INT for latitude, longitude, altitude, velocity, and heading,
- GPS_RAW_INT for GPS fix quality and satellite-related information,
- ATTITUDE for roll, pitch, yaw, and angular rates,
- VIBRATION for vibration levels and clipping counts,
- ESTIMATOR_STATUS for EKF innovation and consistency ratios,

- HEARTBEAT for mode and armed state,
- SYS_STATUS for battery and system health,
- VFR_HUD for speed-related navigation information.

This message set supports the transition from GPS-only anomaly detection toward multi-sensor cross-validation, which is central to RQ2 and RQ3 [11].

Sampling Strategy

The current methodology uses a sampling rate of 10 Hz. This is appropriate because it provides sufficient temporal resolution for short-term behavioural analysis while remaining computationally manageable for real-time processing [11]. At this rate, a 30-sample window corresponds to approximately 3 seconds of telemetry, which is consistent with the project goal of near-real-time spoofing detection.

Threat Model and Scenario Types

The threat model includes several spoofing patterns:

- gradual position drift attacks,
- sudden position jump attacks,
- manipulated velocity or trajectory inconsistency,
- broader GPS takeover scenarios.

These scenarios are generated only in controlled simulation, consistent with the project's defensive and safety-oriented scope [11, 12]. The methodology also anticipates varying attack intensity to support the later breaking-point study, where the operational limit of the detector will be measured rather than assumed [12].

Data Processing and Dataset Construction

Cleaning and Validation

The data processing pipeline begins by cleaning raw telemetry logs. This includes numeric type conversion, removal of incomplete or invalid rows, filtering for acceptable GPS fix conditions, connection-state validation, duplicate removal, stale data detection, and removal of large temporal gaps [11]. These steps are necessary to prevent noise, disconnection artefacts, and malformed records from distorting the training and evaluation process.

Automated Labelling

The methodology uses a rule-based auto-labelling stage to assign row-level anomaly labels before machine learning training. This weakly supervised approach is suitable because manually labelling all telemetry records across many logs would be impractical and inconsistent [11]. The heuristics cover:

- position jumps,
- abnormal speed changes,
- GPS quality degradation,
- stale signal behaviour,
- IMU-related anomalies,
- estimator innovation spikes,
- GPS–IMU consistency violations,
- mode or failsafe anomalies.

The methodology uses these labels as an intermediate supervisory signal rather than treating them as perfect ground truth. This is important methodologically because heuristic labels can contain bias, so later evaluation must consider both classifier performance and the limitations of the labelling process itself.

Sliding Window Dataset Construction

After row-level processing, the telemetry is segmented into overlapping windows for model training and inference. The current configuration uses:

- window length: 30 samples,
- stride: 15 samples,
- overlap: 50%,
- window label: anomalous if more than 50% of rows in the window are labelled anomalous.

This design is appropriate because spoofing often manifests over short temporal patterns rather than isolated single records. The windowing strategy also aligns with the current live inference buffer design, which supports methodological consistency between offline training and online deployment [11, 12].

Dataset Status

The current methodology documentation reports a dataset built from sliding telemetry windows with both normal and anomalous examples, using 34 features across GPS, IMU-related, estimator, and health signals, with separate train, validation, and test partitions [11]. Earlier checkpoint material shows that the project first progressed through a smaller baseline dataset and then planned expansion toward richer cross-sensor integration and broader scenario diversity [12]. This staged evolution is methodologically acceptable because the research is explicitly iterative and still in progress.

Feature Engineering and Sensor Cross-Validation

Feature Categories

The methodology uses four main feature groups:

1. **GPS features:** local position, altitude, relative altitude, ground speed, heading, fix type, satellite count, EPH, and EPV.
2. **IMU and vibration features:** attitude angles, angular rates, vibration measures, and clipping indicators.
3. **Estimator features:** innovation residuals and consistency ratios from the EKF.
4. **Health features:** battery state, armed state, failsafe state, connection state, and stale-data indicators.

This structure supports both RQ2 and RQ3 because it captures not only navigation outputs but also the degree to which those outputs are physically and estimator-wise consistent with the robot's internal state [11].

Core Cross-Validation Principle

The central methodological idea is that spoofed GPS motion should be detectable when it becomes inconsistent with the robot's own physical motion and state estimates. For example, a wind disturbance should affect both motion-related GPS readings and body dynamics, whereas certain spoofing patterns may change GPS-derived motion without corresponding IMU or estimator behaviour. This cross-validation principle is the key move beyond single-sensor anomaly detection.

Feature Transformation

GPS coordinates are transformed into a local coordinate frame for motion-based analysis, enabling metric comparisons across time. Standardisation is applied where required using

scalers fit only on training data, which helps avoid information leakage and supports more valid evaluation.

Machine Learning Methodology

Baseline and Final Model Strategy

The project uses a staged modelling strategy. A baseline model is first established using telemetry-derived tabular features and supervised classification. The broader final methodology then extends this baseline with stronger cross-sensor validation, trust assessment, and breaking-point evaluation [12].

Models Considered

Current project materials identify two main model families:

- **Random Forest**, used as the primary model because it is fast, interpretable, and suitable for tabular features.
- **1D CNN or related temporal models**, explored as an alternative for capturing sequential patterns.

The choice of Random Forest as the main current model is justified by its low inference cost, resistance to moderate data imbalance, compatibility with flattened sliding windows, and suitability for near-real-time operation. Deep learning remains relevant, but in this methodology it is treated as a comparative or future extension rather than the sole main method.

Training Procedure

The current documented training setup uses train, validation, and test splits, with feature windows flattened or otherwise prepared according to model type. The methodology requires that all preprocessing steps be applied consistently across training and live inference, including scaling, window length, and feature ordering, to ensure that evaluation results are meaningful and reproducible.

Evaluation Metrics

The methodology evaluates model performance using standard metrics:

- accuracy,
- precision,
- recall,
- F1-score,

- confusion matrix analysis,
- latency or inference-time measurements.

These metrics are necessary because a useful detector must not only identify attacks correctly but also limit false positives and operate within practical timing constraints. The methodology therefore treats real-time performance as a first-class evaluation dimension, not merely a secondary implementation detail.

Breaking-Point Analysis

A distinctive part of this methodology is the planned breaking-point analysis, identified in supervisory guidance as the project's key novel contribution [12]. In this stage, spoofing attack intensity, drift rate, or offset magnitude will be systematically varied to determine the point at which classifier performance begins to degrade below acceptable operational thresholds.

This is important because high detection accuracy on a limited test set does not reveal the operational boundary of the system. The breaking-point study is therefore designed to convert the detector from a simple classifier into a quantitatively characterised defensive system with a known effective range.

Dynamic GPS Trust Index Methodology

Motivation

Most spoofing detection approaches produce binary outputs, such as spoofed or authentic, yet autonomous robotic systems often require a more graded judgement about whether GPS should be fully trusted, partially trusted, or temporarily down-weighted [9, 10]. This motivates the Dynamic GPS Trust Index in RQ3.

Trust Index Concept

The methodology defines the Dynamic GPS Trust Index as a continuous reliability score $T(t)$ in the interval $[0, 1]$, where higher values indicate stronger confidence in GPS reliability and lower values indicate increased suspicion or degraded trust. Rather than replacing anomaly detection, the trust index interprets anomaly evidence over time and provides a smoother input for later navigation decision logic.

Planned Formulation

The trust score is planned to be updated over time using a temporally smoothed function of anomaly probability:

$$T(t) = \alpha T(t-1) + (1-\alpha)(1-p_{\text{anom}}(t))$$

where $p_{\text{anom}}(t)$ is the anomaly probability produced by the current model and α is a smoothing parameter controlling trust persistence. This formulation is consistent with the project's earlier methodology direction and with the literature's broader movement from binary detection to continuous trust assessment [9, 10, 13].

Methodological Role of the Trust Index

The trust index serves three purposes in the methodology:

1. it links detection output to operational interpretation,
2. it reduces abrupt switching caused by momentary noise or isolated false alarms,
3. it supports future integration with navigation reweighting or failsafe logic.

The trust mechanism will therefore be evaluated not only conceptually but also in terms of responsiveness, stability, and sensitivity under varied spoofing scenarios.

Research Plan and Timeline

Overview

The research plan is structured as a focused trimester-based program that progresses from research initiation and literature consolidation to dataset preparation, model refinement, experimental evaluation, thesis writing, and final dissemination preparation. This plan is designed to reflect a scholarly research workflow milestones.

The timeline is organised to ensure that each stage of the project contributes directly to the research questions, methodological development, implementation maturity, and evaluation outcomes. It also supports a publishable research direction by emphasising reproducibility, technical validation, analytical clarity, and coherent written reporting.

Phase Plan

Phase 1: Research Framing and Methodology Consolidation

Period: Early March 2026

- Confirm the final research scope, problem framing, and alignment with the research questions.
- Consolidate literature relevant to GPS spoofing, sensor fusion, anomaly detection, and trust modelling.

- Refine the methodological structure and experimental direction of the project.

Output: Consolidated research scope and formalised methodology foundation.

Phase 2: Literature Integration and Evaluation Design

Period: Mid to late March 2026

- Strengthen the literature-guided justification for the chosen methodology.
- Define the evaluation strategy, including classification metrics, latency assessment, and breaking-point analysis direction.
- Refine the mapping between research objectives, features, models, and expected outputs.

Output: Literature-informed methodological justification and evaluation framework.

Phase 3: Data and Artefact Consolidation

Period: April 2026

- Consolidate telemetry logs, preprocessing steps, feature engineering, and dataset generation artefacts.
- Verify the quality of automated labelling and improve consistency across experimental scenarios.
- Prepare reproducible research artefacts to support later training, validation, and reporting.

Output: Cleaned and reproducible labelled dataset with supporting implementation artefacts.

Phase 4: Model Refinement and Thesis Draft Development

Period: Late April 2026

- Refine baseline and improved anomaly detection models using structured telemetry features.
- Strengthen cross-sensor validation logic and prepare initial experimental results.
- Integrate methodology, system design, and preliminary findings into the thesis draft.

Output: Refined detection framework and developed thesis draft.

Phase 5: Review, Validation, and Research Synthesis

Period: Early to mid May 2026

- Review the overall research contribution, analytical consistency, and experimental validity.

- Finalise performance analysis, including detector behaviour, trust assessment, and evaluation interpretation.
- Revise the written document to improve coherence, rigour, and contribution clarity.

Output: Revised analytical results and near-final research manuscript.

Phase 6: Defence and Dissemination Preparation

Period: Mid to late May 2026

- Prepare the final research presentation and defence materials.
- Finalise the thesis document, figures, tables, and references.
- Prepare dissemination-ready outputs suitable for presentation, portfolio inclusion, and potential future publication development.

Output: Final defended thesis and dissemination-ready research package.

Timeline Table

Table 2: Trimester 1, 2026 Research Timeline and Outputs		
Phase	Period	Activities / Main Output
1	Early March 2026	Research initiation, confirmation of scope, refinement of research questions, and methodology consolidation.
2	Mid–Late March 2026	Literature integration, methodological justification, and evaluation framework design.
3	April 2026	Telemetry consolidation, preprocessing refinement, artefact preparation, and labelled dataset validation.
4	Late April 2026	Model refinement, cross-sensor detection development, preliminary result generation, and thesis draft writing.
5	Early–Mid May 2026	Experimental review, analytical refinement, trust and performance interpretation, and manuscript revision.
6	Mid–Late May 2026	Defence preparation, final thesis integration, and dissemination-ready research output preparation.

Planned Outputs Across the Research Period

The plan produces a sequence of research-oriented outputs that are appropriate for scholarly reporting. Initial stages establish the conceptual and methodological basis of the study, middle stages focus on artefact and model development, and later stages concentrate on validation, interpretation, writing, and dissemination preparation.

This structure is better suited to a publishable research narrative because it emphasises technical maturity, coherent experimentation, reproducibility, and contribution development rather than

administrative course tasks. It also aligns more naturally with thesis writing and potential future paper preparation.

Risk and Delay Management

The timeline allows for risks such as simulation instability, limited dataset diversity, imperfect heuristic labelling, delays in cross-sensor implementation, and the need for iterative revision after evaluation. These risks are managed through a reproducible simulation pipeline, retention of a baseline detection model, staged validation, and progressive integration of writing with implementation and experimentation.

This approach supports research continuity even if some advanced components require further refinement. It also ensures that the project remains methodologically credible by maintaining measurable outputs at each stage of the research period.

Ethical Considerations

Dual-Use Research

GPS spoofing research has an unavoidable dual-use dimension because understanding attack mechanisms can potentially aid misuse if presented irresponsibly. This project addresses that risk by focusing exclusively on detection, resilience, and defensive system design, and by restricting experiments to simulation environments rather than offensive real-world RF deployment [12, 13].

Safety Considerations

The methodology is designed to avoid risks to people, aircraft, public infrastructure, and legitimate navigation systems. All core attack testing is performed in simulation, and no real-world broadcast spoofing is required for the current research plan [8, 12]. This makes simulation not only methodologically useful but also ethically necessary.

Privacy and Confidentiality

The project does not involve human participants or personal data in its main form. The primary data consists of simulated robot telemetry, system logs, and model outputs [12]. Even so, project files, logs, and experimental outputs should be stored securely, versioned responsibly, and managed in line with Deakin University's expectations for research data handling.

Informed Consent and Participant Risk

Because the current methodology does not involve human subjects, informed consent is not presently required in the same way as for human-participant studies. If the scope later expands

to human-operated field trials, pilot interaction, or user studies, appropriate consent and ethics procedures would be added before such research begins.

Ethical Approval and Permissions

At the current simulation-focused stage, formal human research ethics clearance is unlikely to be required because the project does not involve identifiable human data or interventions with participants. However, any move toward hardware testing with live RF behaviour, public-space trials, or human participants would require consultation with the relevant Deakin ethics or safety bodies before proceeding [13].

Responsible Disclosure

If the research uncovers practically significant vulnerabilities in deployed toolchains, autopilot workflows, or telemetry assumptions, those findings will be communicated responsibly through supervisory and academic channels before any wider public release.

Milestones and Progress Tracking

Key Milestones

Progress Monitoring

Progress will be tracked through weekly or regular measurable indicators, including:

- number of usable telemetry logs collected,
- number and diversity of spoofing scenarios represented,
- completion status of preprocessing and feature modules,
- classifier performance metrics,
- measured inference latency,
- completion of trust-index experiments,
- supervisor feedback incorporated into the draft.

SMART Characteristics

This methodology is SMART in the following sense:

- **Specific:** it targets GPS spoofing detection in autonomous mobile robots using PX4-based simulation and telemetry.
- **Measurable:** it uses explicit metrics such as precision, recall, F1-score, latency, and breaking-point thresholds.

Table 3: Research Milestones and Success Criteria

Milestone	Success Criteria	Target Period
Simulation environment operational	PX4 SITL, Gazebo, and MAVLink data streaming working reliably	Achieved / maintained
Telemetry pipeline complete	GPS and related MAVLink telemetry logged and processed consistently	Achieved / refined
Labelled dataset established	Cleaned, labelled, and windowed dataset available for training	Achieved / expanding
Baseline model operational	Working anomaly detector trained and tested on prepared windows	Achieved
Cross-sensor validation complete	GPS consistency logic implemented with non-GPS telemetry support	May–Jun 2026
Threshold and latency analysis complete	Detection threshold justified and runtime measured	Jun 2026
Breaking-point analysis complete	Detection degradation characterised against attack intensity	Jul 2026
Trust index evaluation complete	Continuous GPS reliability score validated in experiments	Jul 2026
Thesis/report draft complete	Methodology, results, and discussion chapters assembled	Aug 2026

- **Achievable:** it builds directly on a working implementation and documented project progress.
- **Relevant:** it addresses a real cybersecurity threat in autonomous systems.
- **Time-bound:** it includes phased deadlines and milestone targets through the trimester research cycle.

Sustainability Measures

Technical Sustainability

The project is designed to be reproducible and maintainable through a configuration-driven pipeline, documented preprocessing steps, scripted dataset creation, and modular live inference components. This makes the research easier to re-run, audit, and extend beyond the immediate assignment.

Data and Code Sustainability

Where permitted, cleaned non-sensitive simulation datasets, configuration files, and code modules can be archived and documented so that future work can build on the same pipeline. This supports long-term research continuity and improves transparency in experimental evaluation.

Academic Sustainability

The project is intended to contribute not only to unit assessment but also to future thesis chapters, possible paper development, and broader research on secure autonomous systems. Earlier supervisory discussions already framed the work as having publication potential if the results are strong and the contribution is clearly articulated [12].

Dissemination Strategy

Findings will be disseminated through:

- unit submissions and thesis chapters,
- supervisor review meetings,
- diagrams, evaluation tables, and reproducible methodological descriptions,
- possible later journal or conference outputs if the results mature sufficiently.

Long-Term Relevance

The long-term relevance of this research lies in the growing deployment of autonomous robots in logistics, inspection, mapping, and security contexts where GNSS dependence remains common. A lightweight, reproducible, telemetry-level spoofing detection framework therefore has continuing relevance even beyond the immediate project scope [3, 4].

Expected Contributions

Methodological Contributions

This research is expected to contribute:

1. a formalised methodology for telemetry-based GPS spoofing detection in autonomous mobile robots,
2. a practical cross-sensor validation framework using GPS, IMU-related, estimator, and health signals,
3. a structured method for evaluating spoofing detection under controlled breaking-point experiments.

Practical Contributions

The project is also expected to contribute:

1. a reproducible PX4–Gazebo experimental pipeline,
2. a near-real-time detection prototype using MAVLink telemetry,
3. a Dynamic GPS Trust Index design for reliability-aware navigation reasoning.

Scientific Contributions

Scientifically, the project aims to contribute:

1. clearer analysis of GPS vulnerabilities in mobile robotic platforms,
2. empirical evidence on the feasibility of real-time sensor fusion spoofing detection under practical constraints,
3. a more nuanced understanding of how spoofing detection degrades as attack strength increases.

Conclusion

This research methodology presents a complete and formal draft of the research design for investigating GPS spoofing detection in autonomous mobile robots using sensor cross-validation, machine learning, and trust-based interpretation. It extends the work initiated in SIT723 and aligns the ongoing implementation with a clear academic methodology suitable for SIT724/SIT746 [11–13].

The methodology is appropriate because it combines controlled simulation, telemetry-driven data collection, structured preprocessing, model-based anomaly detection, real-time evaluation, ethical safeguards, and a realistic project timeline. It is also well aligned to the three research questions by linking vulnerability analysis to attack modelling, sensor fusion to real-time detection, and trust modelling to decision-level reliability assessment.

Most importantly, the methodology is designed as a realistic work-in-progress rather than an overclaimed finished system. By explicitly incorporating remaining tasks such as cross-sensor validation, threshold tuning, latency quantification, and breaking-point analysis, it provides a credible and academically rigorous path for completing the research while maintaining SMART planning and strong research integrity.

References

- [1] M. Ahmad, M. A. Farid, S. Ahmed, K. Saeed, M. Asharf, and U. Akhtar, “Impact and detection of GPS spoofing and countermeasures against spoofing,” in *Proc. Int. Conf. Computing, Mathematics and Engineering Technologies (iCoMET)*, 2019, pp. 1–8.
- [2] M. L. Psiaki and T. E. Humphreys, “GNSS spoofing and detection,” *Proceedings of the IEEE*, vol. 104, no. 6, pp. 1258–1270, 2016.
- [3] E. Kadena, N. H. P. Dai, and L. Ruiz, “Mobile robots: An overview of data and security,” in *Proc. 7th Int. Conf. Information Systems Security and Privacy (ICISSP)*, 2021, pp. 291–299.
- [4] J.-P. A. Yaacoub, H. N. Noura, O. Salman, and A. Chehab, “Robotics cyber security: Vulnerabilities, attacks, countermeasures, and recommendations,” *International Journal of Information Security*, vol. 21, pp. 115–158, 2022.
- [5] S. Dasgupta, M. Rahman, M. Islam, and M. Chowdhury, “A sensor fusion-based GNSS spoofing attack detection framework for autonomous vehicles,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 11, pp. 19255–19268, 2022.
- [6] Z. Wang, Y. Wu, and Q. Niu, “Multi-sensor fusion in automated driving: A survey,” *IEEE Access*, vol. 8, pp. 2847–2868, 2020.
- [7] U. Patil, A. Gunasekaran, R. Bobba, and H. Abbas, “ROS2-based simulation framework for cyberphysical security analysis of UAVs,” *arXiv preprint arXiv:2410.03971*, 2024.
- [8] I. Pekaric, D. Arnold, and M. Felderer, “Simulation of sensor spoofing attacks on unmanned aerial vehicles using the Gazebo simulator,” *arXiv preprint arXiv:2309.09648*, 2023.
- [9] H. Azevedo-Sa, S. K. Jayaraman, C. T. Esterwood, X. J. Yang, L. P. Robert Jr., and D. M. Tilbury, “Real-time estimation of drivers’ trust in automated driving systems,” *International Journal of Social Robotics*, vol. 13, pp. 1911–1933, 2021.
- [10] U. Tariq and B. Tariq, “Signal characteristic analysis and anomaly detection for GPS spoofing mitigation,” *Ubiquitous Technology Journal*, vol. 1, no. 1, pp. 10–22, 2025.
- [11] N. Sarwara, “GPS Spoofing Detection Methodology,” project methodology notes and pipeline documentation, 2026.
- [12] N. Sarwara, “SIT724/SIT746 Research Progress Checkpoint 2 (Task 3.2P),” Deakin University submission, Apr. 2026.
- [13] N. Sarwara, “Research Methodology for Detection and Simulation of GPS Spoofing in Mobile Robots Using Sensor Cross-Validation,” SIT723 Research Methodology draft, Sep. 2025.